Korean IGT – Experimental Plan & Baselines

Bolmo 1B | Zero-shot Prompting | LoRA Fine-tuning | Evaluation Strategy

Team Experimental Protocol – Version 1.1 | 12 November 2026

Team Members: Soyeong and project team

Status: Data validation and experimental pipeline preparation

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1. Project Objective

The objective of this project is to investigate whether Bolmo 1B can generate accurate interlinear glossed text (IGT) from Korean sentences and whether LoRA fine-tuning improves its performance compared with zero-shot prompting.

We will investigate three main research questions:

RQ1. Does step-by-step prompting outperform all-in-one prompting for Korean IGT generation?

RQ2. Does LoRA fine-tuning improve morphological gloss accuracy compared with zero-shot prompting?

RQ3. How do different LoRA hyperparameters affect performance, and how does checkpoint selection based on validation loss compare with selection based on chrF++?

Our experiments will focus initially on Korean as a controlled experimental language. Korean is not generally considered a low-resource language, so the findings will not automatically generalize to low-resource languages. Extending the experiments to other languages may be considered after establishing a reliable baseline.

1.1 General Setup

Component	Description
Base model	allenai/Bolmo-1B
Language	Korean
Dataset	Korean IGT dataset from the lowresource-igt Hugging Face collection

Component	Description
Input	Original Korean text and English translation
Intermediate output	Transliteration or transcription, depending on dataset verification
Final output	Morphological gloss
Fine-tuning method	LoRA (Parameter-Efficient Fine-Tuning)
Evaluation	Morpheme-level accuracy, word-level accuracy, chrF++, validation loss
Data splitting	Fixed training, validation, and test sets
Computing environment	Google Colab with GPU

The input and output specifications follow the tutor's proposed experimental direction. However, they remain subject to dataset verification, particularly the accuracy of the English translations and the meaning of the transcription field.

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2. Current Situation and Identified Problems

2.1 Dataset Quality Issues

During our initial inspection of the Korean IGT dataset, we identified several potential data quality issues.

These include:

- Korean sentences that appear unnatural or grammatically questionable.
- English translations that do not accurately correspond to the Korean source sentences.
- Possible inconsistencies between source text, transcription, and morphological gloss annotations.
- Uncertainty about whether the dataset's `transcription` field represents transliteration, phonetic transcription, or another linguistic annotation format.

Some unusual sentences may originate from linguistic examples designed to illustrate specific grammatical phenomena. Therefore, grammatical unusualness alone should not be treated as evidence of incorrect data.

We must distinguish genuine annotation errors from intentional linguistic examples before removing or modifying records.

2.2 Impact on the Experiments

These issues may affect the reliability of both training and evaluation.

For example:

- Incorrect translations may mislead the model when translation is included in the input.
- Incorrect gloss annotations may cause the model to learn unreliable patterns.
- Misaligned transcription and gloss labels may introduce errors into the two-step pipeline.
- Incorrect gold labels may produce misleading evaluation scores.

As a result, we should not conduct the final LoRA fine-tuning experiments or report final evaluation scores until the dataset has been sufficiently verified.

2.3 Current Decision

We will divide the work into two parallel activities:

1. **Dataset validation and cleaning**, led by Soyeong.
2. **Experimental pipeline development**, carried out by the remaining team members.

While the dataset is being reviewed, other team members can prepare and test their code using a small sample.

These preliminary tests are intended to verify that the code works, not to measure final model performance.

The final experiments will begin after the dataset and experimental conditions have been agreed upon.

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3. Team Responsibilities and Parallel Workflow

3.1 Soyeong – Korean Dataset Validation and Cleaning

Soyeong will lead the inspection and preparation of the Korean dataset.

Tasks:

1. Load and inspect the original Korean IGT dataset.
2. Examine the available fields, including original text, transcription, glosses, and English translation.
3. Verify whether Korean sentences correspond to their English translations.
4. Check whether transcription and gloss annotations are aligned with the source text.
5. Identify missing values, duplicated records, inconsistent annotations, and potential errors.
6. Determine whether the transcription field represents transliteration or another annotation type.
7. Document the identified problems and define transparent cleaning criteria.
8. Flag uncertain cases for discussion with the team or tutor.
9. Prepare the verified dataset and a reproducible cleaning script.
10. Establish fixed training, validation, and test splits after the cleaning criteria are finalized.

Important considerations:

- Do not automatically delete sentences simply because they appear unnatural.
- Preserve the original dataset.
- Record every filtering or correction decision.
- Do not silently rewrite linguistic annotations without sufficient justification.
- Keep track of excluded records and the reasons for exclusion.
- Check for duplicate or near-duplicate examples across the final data splits.

Deliverables:

- Dataset quality assessment.
- Cleaning and validation script.
- Documented cleaning criteria and error log.
- Verified Korean IGT dataset.
- Fixed train/validation/test split manifest.

3.2 Team Member A – Zero-shot Prompting Pipeline

While Soyeong validates the dataset, one team member will develop the zero-shot inference pipeline.

Tasks:

1. Set up the Python and Hugging Face environment.
2. Load the pretrained Bolmo 1B model.
3. Implement all-in-one zero-shot prompting.
4. Implement step-by-step zero-shot prompting.
5. Prepare input formatting for original text and English translation.
6. Implement automatic processing of multiple dataset records.
7. Extract the generated transliteration/transcription and gloss.
8. Save raw model outputs and parsed predictions.
9. Conduct small-scale smoke tests using approximately 5–10 examples.
10. Ensure both prompting methods use consistent generation settings.

Deliverables:

- Zero-shot inference script or notebook.
- All-in-one and step-by-step prompt templates.
- Output parsing and prediction-saving functions.
- Preliminary smoke-test results.

These smoke tests are not part of the final reported evaluation.

3.3 Team Member B – LoRA Training and Evaluation Pipeline

Another team member will prepare the fine-tuning and evaluation infrastructure.

Tasks:

1. Verify which Bolmo 1B modules are compatible with LoRA.
2. Implement LoRA adapter initialization.
3. Prepare the training data formatting and tokenization pipeline.
4. Ensure that the loss is calculated on the intended target response tokens.
5. Implement training and checkpoint saving.

6. Test a small number of training steps using sample records.
7. Verify that saved LoRA adapters can be reloaded.
8. Implement validation loss calculation.
9. Implement chrF++ evaluation.
10. Implement morpheme-level and word-level accuracy following Ginn et al. (2024).
11. Prepare scripts for comparing different LoRA configurations.

Deliverables:

- LoRA training script or notebook.
- Baseline LoRA configuration.
- Checkpoint saving and loading functionality.
- Evaluation script.
- Preliminary smoke-test results.

The exact implementation of word-level and morpheme-level accuracy must be verified against the referenced paper before final evaluation.

3.4 Team Coordination

The proposed responsibilities are intended to allow parallel progress.

All team members should agree on:

- Dataset column names and input/output formats.
- Prompt templates and output formats.
- Shared model and package versions.
- Evaluation rules.
- File naming conventions.
- Train/validation/test split identifiers.
- The criteria for considering the dataset ready for final experiments.

The dataset validation process should remain collaborative. Difficult linguistic cases should be discussed rather than resolved by one person without review.

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4. Preliminary Pipeline Testing During Data Cleaning

The purpose of preliminary testing is to ensure that the experimental code works before the final dataset becomes available.

4.1 Small Test Sample

We will initially use approximately 5–10 records to test the pipelines.

These records may be drawn from the original dataset or constructed specifically for debugging.

They must be clearly labeled as development examples.

4.2 Zero-shot Smoke Test

The zero-shot pipeline should be able to:

1. Load Bolmo 1B successfully.
2. Accept the required input fields.
3. Generate an output for a single example.
4. Process multiple examples automatically.
5. Run both prompting strategies.
6. Save raw and parsed predictions.
7. Handle malformed or unexpected model outputs.

4.3 LoRA Smoke Test

The LoRA pipeline should be able to:

1. Load the pretrained model.
2. Attach compatible LoRA adapters.
3. Prepare tokenized training examples.
4. Complete a small number of training steps.
5. Calculate training loss.
6. Save a checkpoint.
7. Reload the saved adapter.

8. Generate an output using the reloaded adapter.

A successful smoke test does not require high accuracy or a consistently decreasing loss.

4.4 Evaluation Smoke Test

The evaluation pipeline should be tested using a small set of known reference and prediction pairs.

The tests should verify:

- Correct calculation of chrF++.
- Correct implementation of word-level accuracy.
- Correct implementation of morpheme-level accuracy.
- Appropriate handling of missing or malformed predictions.
- Appropriate handling of different prediction and reference lengths.
- Consistent result formatting.

4.5 Transition to Final Experiments

Once the dataset has been verified:

1. Freeze the dataset version.
2. Fix the training, validation, and test splits.
3. Confirm the final input and target fields.
4. Replace the preliminary test data with the verified dataset.
5. Re-run the complete experimental pipelines.
6. Use only the final experimental runs for the main research results.

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5. Zero-shot Experiments

We will evaluate the unmodified pretrained Bolmo 1B model using two prompting strategies.

Both strategies will use:

- The same held-out test examples.
- The same base model.
- The same verified input fields.
- The same deterministic generation settings where supported.
- The same evaluation script.

5.1 Experiment A – All-in-One Prompting

The model will predict the transliteration and morphological gloss in a single call.

Prompt template:

You are a linguist specializing in interlinear glossed text (IGT).

Task: Given a Korean sentence and its English translation, predict its transliteration and morphological gloss.

Language: Korean

Original text: {text}

English translation: {translation}

Instructions:

1. Predict the transliteration of the original Korean text.
2. Predict the corresponding morphological gloss.
3. Preserve the original word and morpheme order.
4. Use standard linguistic glossing abbreviations.
5. Do not provide explanations or additional commentary.

Output format:

Transliteration: ...

Gloss: ...

Procedure:

1. Load the pretrained Bolmo 1B model.
2. Load the verified Korean test set.

3. Insert the source text and translation into the prompt.
4. Generate both outputs in one model call.
5. Save raw outputs and parsed predictions.
6. Evaluate the generated glosses.

5.2 Experiment B – Step-by-Step Prompting

The model will generate the transliteration first and then use its prediction to generate the gloss.

Step 1 – Predict Transliteration

You are a linguist specializing in Korean language annotation.

Task: Predict the transliteration of the following Korean text.

Language: Korean

Original text: {text}

English translation: {translation}

Instructions:

1. Transliterate the original text.
2. Preserve the original word order.
3. Do not generate morphological glosses.
4. Do not provide explanations.

Output only the transliteration:

The generated result will be stored as `{predicted_transliteration}`.

Step 2 – Predict Morphological Gloss

You are a linguist specializing in interlinear glossed text (IGT).

Task: Predict the morphological gloss using the original Korean text, its English translation, and the predicted transliteration.

Language: Korean

Original text: {text}

English translation: {translation}

Predicted transliteration: {predicted_transliteration}

Instructions:

1. Predict the corresponding morphological gloss.
2. Preserve the original word and morpheme order.
3. Use standard linguistic glossing abbreviations.
4. Do not provide explanations.

Output only the gloss:

Mandatory fairness rule:

During test inference, Step 2 must use the model's predicted transliteration, never the gold transliteration.

5.3 Zero-shot Baseline Settings

Parameter	Setting
Model	allenai/Bolmo-1B
Fine-tuning	None
Sampling	do_sample=False
Prompt language	English
Input	Korean text + verified English translation
Evaluation dataset	Identical held-out Korean test split
Primary output	Morphological gloss

The prompt terminology will be updated if the dataset field is confirmed to represent transcription rather than transliteration.

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6. LoRA Fine-tuning

We will fine-tune Bolmo 1B using LoRA while keeping the pretrained base model weights frozen.

The initial objective is to establish a working baseline and subsequently compare different hyperparameter configurations.

6.1 Baseline Hyperparameters

Parameter	Baseline
Base model	allenai/Bolmo-1B
LoRA rank (r)	16
LoRA alpha	32
LoRA dropout	0.05
Learning rate	2e-4
Epochs	3
Random seed	42
Batch size	Determine based on GPU memory
Checkpoint saving	After each epoch

These are proposed initial settings, not validated optimal values.

6.2 LoRA Hyperparameter Experiments

We will investigate how different LoRA configurations affect model performance.

Experiment	Rank (r)	Alpha	Learning Rate
A – Baseline	16	32	2e-4
B – Lower LR	16	32	1e-4
C – Lower Rank	8	16	2e-4
D – Higher Rank	32	64	2e-4

All experiments will use the same verified dataset splits, three training epochs, dropout rate of 0.05, and random seed of 42.

Comparisons:

- A vs. B: Effect of changing the learning rate.
- A vs. C vs. D: Effect of changing rank and alpha configurations.

Since rank and alpha change together in C and D, the individual effect of rank cannot be isolated.

If computing resources are limited, Experiments A and B will be prioritized, with C and D treated as optional extensions.

6.3 Training Task 1 – Transliteration

Input:

- Original Korean text.
- Verified English translation.

Target:

- Gold transliteration.

Prompt template:

You are a linguist specializing in Korean.

Task: Transliterate the following Korean text.

Original text: {text}

English translation: {translation}

Output only the transliteration:

Target response: `{gold_transliteration}`

6.4 Training Task 2 – Morphological Glossing

Input:

- Original Korean text.
- Verified English translation.
- Gold transliteration.

Target:

- Gold morphological gloss.

Prompt template:

You are a linguist specializing in interlinear glossed text (IGT).

Task: Predict the morphological gloss of the following Korean text.

Original text: {text}

English translation: {translation}

Transliteration: {gold_transliteration}

Output only the morphological gloss:

Target response: `{gold_gloss}`

We will initially use one shared LoRA adapter for both training tasks.

Gold transliteration is permitted during supervised training. During validation and test generation, the predicted transliteration must be used.

This creates a difference between the training and inference conditions, which may lead to error propagation. We will document this limitation and examine its effect during error analysis.

6.5 Fine-tuning Procedure

1. Complete dataset validation and freeze the experimental splits.
2. Construct two task-specific training records for each valid example.
3. Verify LoRA-compatible modules in the Bolmo architecture.
4. Initialize a fresh adapter for each hyperparameter configuration.
5. Train each configuration on the same training dataset.
6. Save checkpoints after each epoch.
7. Compute validation loss and generate validation predictions.
8. Calculate chrF++ on the generated validation glosses.
9. Compare the different hyperparameter configurations using validation results.
10. Select checkpoints using predefined selection criteria.
11. Evaluate the selected models on the held-out test set.

The test set must not be used to select hyperparameters or checkpoints.

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7. Evaluation Strategy

7.1 Final Reporting Metrics

Metric	Interpretation
Morpheme-level accuracy	Proportion of correctly predicted morpheme gloss labels
Word-level accuracy	Proportion of words with entirely correct glosses
chrF++	Character- and word-n-gram similarity between predicted and reference glosses
Validation loss	Model loss on held-out validation examples

The exact alignment and matching rules for morpheme-level and word-level accuracy will follow Ginn et al. (2024), subject to verification of the original evaluation methodology.

All models will be evaluated using the same implementation.

Predictions must not be manually repaired or selectively omitted after observing the results.

7.2 Checkpoint Selection Mini-experiment

We will compare two checkpoint selection strategies.

Strategy	Selection Rule
A – Validation Loss	Select the checkpoint with the lowest validation loss
B – Validation chrF++	Select the checkpoint with the highest validation chrF++

For chrF++, the model will generate validation glosses using the complete step-by-step pipeline.

The selected checkpoints will then be evaluated on the same held-out test set.

If both strategies select the same checkpoint, this agreement will be reported.

The comparison will be conducted using predefined evaluation procedures to avoid selecting models based on test results.

7.3 Error Analysis

We will examine representative errors, including:

- Incorrect transliteration or transcription.

- Incorrect morphological gloss labels.
- Missing or additional morphemes.
- Word or morpheme alignment errors.
- Errors propagated from the first prediction step.
- Cases where all-in-one and step-by-step prompting produce different results.

We will also consider whether specific dataset characteristics are associated with recurring errors.

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8. End-to-End Workflow and Timeline

The workflow is organized into phases rather than fixed calendar deadlines. Progression to the final experiments depends on successful dataset validation.

Phase 1 – Parallel Preparation

Soyeong:

- Inspect and validate the Korean dataset.
- Identify annotation and translation errors.
- Establish cleaning criteria.
- Prepare the cleaned dataset.

Other team members:

- Develop zero-shot prompting code.
- Develop LoRA training code.
- Develop evaluation functions.
- Conduct small smoke tests.
- Resolve technical issues in advance.

Expected outcome:

A validated dataset and working experimental pipelines.

Phase 2 – Dataset Freeze and Integration

Once dataset validation is complete:

1. Agree on the final dataset version.
2. Confirm the input and output fields.
3. Fix train/validation/test splits.
4. Connect the verified dataset to all experimental pipelines.
5. Run a complete end-to-end integration test.

Expected outcome:

A reproducible experimental setup ready for final runs.

Phase 3 – Zero-shot Baseline Experiments

1. Run all-in-one prompting on the held-out test set.
2. Run step-by-step prompting on the same test set.
3. Save all predictions.
4. Calculate final evaluation metrics.
5. Compare the two zero-shot strategies.

Expected outcome:

Zero-shot baseline results.

Phase 4 – LoRA Fine-tuning Experiments

1. Train the baseline LoRA configuration.
2. Train alternative hyperparameter configurations.
3. Save checkpoints.
4. Evaluate checkpoints on validation data.
5. Compare validation loss and chrF++.

Expected outcome:

Trained LoRA adapters and validation results.

Phase 5 – Model Selection and Final Evaluation

1. Select checkpoints according to the predefined validation criteria.
2. Evaluate selected checkpoints on the held-out test set.
3. Calculate morpheme-level and word-level accuracy.
4. Compare zero-shot and fine-tuned systems.
5. Conduct error analysis.

Expected outcome:

Final experimental results and research findings.

Phase 6 – Reporting

1. Prepare the final comparison tables.
2. Summarize the experimental methodology.
3. Report the dataset quality issues and cleaning decisions.
4. Discuss the results and limitations.
5. Prepare the final report or presentation.

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9. Expected Results Tables

9.1 Main Experiment Comparison

Experiment	Morpheme Accuracy	Word Accuracy
Zero-shot: All-in-one	TBD	TBD
Zero-shot: Step-by-step	TBD	TBD
LoRA: Loss-selected	TBD	TBD
LoRA: chrF++-selected	TBD	TBD

9.2 LoRA Hyperparameter Comparison

Experiment	Best Validation Loss ↓	Best Validation chrF++ ↑
A – Baseline	TBD	TBD
B – Lower Learning Rate	TBD	TBD

Experiment	Best Validation Loss ↓	Best Validation chrF++ ↑
C – Lower Rank	TBD	TBD
D – Higher Rank	TBD	TBD

9.3 Dataset Quality Summary

Item	Result
Original dataset size	TBD
Records reviewed	TBD
Records flagged for potential errors	TBD
Records excluded	TBD
Final verified dataset size	TBD
Training samples	TBD
Validation samples	TBD
Test samples	TBD

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10. Deliverables

Dataset and Documentation

- Dataset quality assessment.
- Cleaning script and documented cleaning criteria.
- Verified Korean IGT dataset.
- Fixed train/validation/test split manifest.

Zero-shot Experiments

- All-in-one prompting script.
- Step-by-step prompting script.
- Raw and parsed model predictions.
- Zero-shot evaluation results.

LoRA Fine-tuning

- LoRA training script.

- Hyperparameter configurations.
- Saved model adapters and checkpoints.
- Training and validation logs.

Evaluation

- Evaluation script.
- Morpheme-level and word-level accuracy results.
- Validation loss and chrF++ comparison.
- Final model comparison tables.
- Error analysis.

Final Research Output

- Experimental methodology.
- Results and discussion.
- Dataset quality limitations.
- Final report or presentation.

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11. Important Decisions and Open Questions

Before finalizing the experiments, the team should confirm the following points.

1. Dataset Quality

Are the original Korean sentences, English translations, transcription fields, and gloss annotations sufficiently reliable for training and evaluation?

2. Translation Input

The tutor's original proposal specifies both the original script and English translation as input.

If the translation field contains substantial errors, should we clean the translations, exclude unreliable records, or investigate a script-only experimental setting with the tutor's approval?

3. Transliteration vs. Transcription

Does the dataset's transcription field represent transliteration, phonetic transcription, or another annotation format?

The prompts, training targets, and evaluation procedures must reflect the actual annotation type.

4. Language Scope

Is Korean sufficient for the initial experiments, or should another language be included to better address the project's low-resource language motivation?

5. Computing Resources

Can the available Google Colab environment support all proposed LoRA experiments, or should the hyperparameter comparison be reduced?

6. Evaluation Implementation

What exact alignment and matching rules are used for morpheme-level and word-level accuracy in Ginn et al. (2024)?

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12. Summary and Immediate Next Steps

Our immediate priority is to establish a reliable dataset while developing the experimental infrastructure in parallel.

Soyeong will focus on inspecting and cleaning the Korean IGT dataset. During this period, the remaining team members will implement and test the zero-shot prompting, LoRA training, and evaluation pipelines using small development samples.

We will not treat preliminary results from unverified data as final experimental findings.

Once the dataset has been validated and the input/output specifications have been confirmed, we will freeze the data splits and execute the final experiments.

The experiments will compare all-in-one and step-by-step zero-shot prompting, zero-shot and LoRA fine-tuning, different LoRA hyperparameter configurations, and checkpoint selection using validation loss versus chrF++.

This workflow will allow the team to make progress in parallel while reducing the risk of training models on unreliable annotations or repeating expensive experiments.

The immediate goal is to prepare a reliable, reproducible experimental setup before running the final model comparisons.